Software Engineering

BCSE301L

WINTER 2024-2025

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PRP-217-4

Course Objectives

- To introduce the essential Software Engineering concepts.
- To impart concepts and skills for performing analysis, design, develop, test and evolve efficient software systems of various disciplines and applications
- To make familiar about engineering practices, standards and metrics for developing software components and products.

Course Outcomes

- Apply and assess the principles of various process models for the software development.
- Demonstrate various software project management activities that include planning , Estimations, Risk assessment and Configuration Management
- Perform Requirements modelling and apply appropriate design and testing heuristics to produce quality software systems.
- Demonstrate the complete Software life cycle activities from requirements analysis to maintenance using the modern tools and techniques.
- Escalate the use of various standards and metrics in evaluating the process and product.

Syllabus

Module:1	**Overview Of Software Engineering**	**6 hours**
Nature of Software, Software Engineering, Software process, project, product, Process Models Classical Evolutionary models, Introduction to Agility - Agile Process-Extreme programming - XP Process – Principles of Agile Software Development framework - Overview of System Engineering		
Module:2	**Introduction To Software Project Management**	**6 hours**
Planning, Scope, Work break-down structure, Milestones, Deliverables, Cost and Estimates - (Human Resources, Time-scale, Costs), Risk Management, RMMM Plan, CASE TOOLS, Agile Project Management, Managing team dynamics and communication, Metrics and Measurement		
Module:3	**Modelling Requirements**	**8 hours**
Software requirements and its types, Requirements Engineering process, Requirement Elicitation, System Modeling – Requirements Specification and Requirement Validation, Requirements Elicitation techniques, Requirements management in Agile.		

Syllabus

Module:4	Software Design	8 hours
Design concepts and principles - Abstraction - Refinement - Modularity Cohesion coupling, Architectural design, Detailed Design Transaction Transformation, Refactoring of designs, Object oriented Design User-Interface Design		
Module:5	**Validation And Verification**	**7 hours**
Strategic Approach to Software Testing, Testing Fundamentals Test Plan, Test Design, Test Execution, Reviews, Inspection and Auditing – Regression Testing – Mutation Testing - Object oriented testing - Testing Web based System - Mobile App testing – Mobile test Automation and tools – DevOps Testing – Cloud and Big Data Testing		

Syllabus

Module:6	**Software Evolution**	**4 hours**
Software Maintenance, Types of Maintenance, - Software Configuration Management – Overview – SCM Tools. Re-Engineering, Reverse Engineering, Software Reuse		
Module:7	**Quality Assurance**	**4 hours**
Product and Process Metrics, Quality Standards Models ISO, TQM, Six-Sigma, Process improvement Models: CMM & CMMI. Quality Control and Quality Assurance - Quality Management - Quality Factors - Methods of Quality Management		
Module:8	**Contemporary Issues**	**2 hours**

Assessment

Assessment Head	Maximum Marks	Weightage	Tentative Date
Digital Assessment 1	10	10	**29-03-2025**
Quiz 1	10	10	**23-01-2025**
Quiz 2	10	10	**11-03-2025**
Continuous Assessment Test 1	50	15	As Per University
Continuous Assessment Test 2	50	15	As Per University
Total	130	60	